

MobiCNV : a simple and fast method to detect Copy Number Variants in Illumina gene panels, clinical exomes and exomes data



Henri Pégeot¹, Charles Van Goethem², Gema Garcia-Garcia¹, Reda Zenagui¹, Delphine Lacourt¹, Olivier Ardouin^{1,2}, Mireille Claustres¹, Michel Koenig¹, Mireille Cossée¹, Anne-Françoise Roux¹, David Baux¹

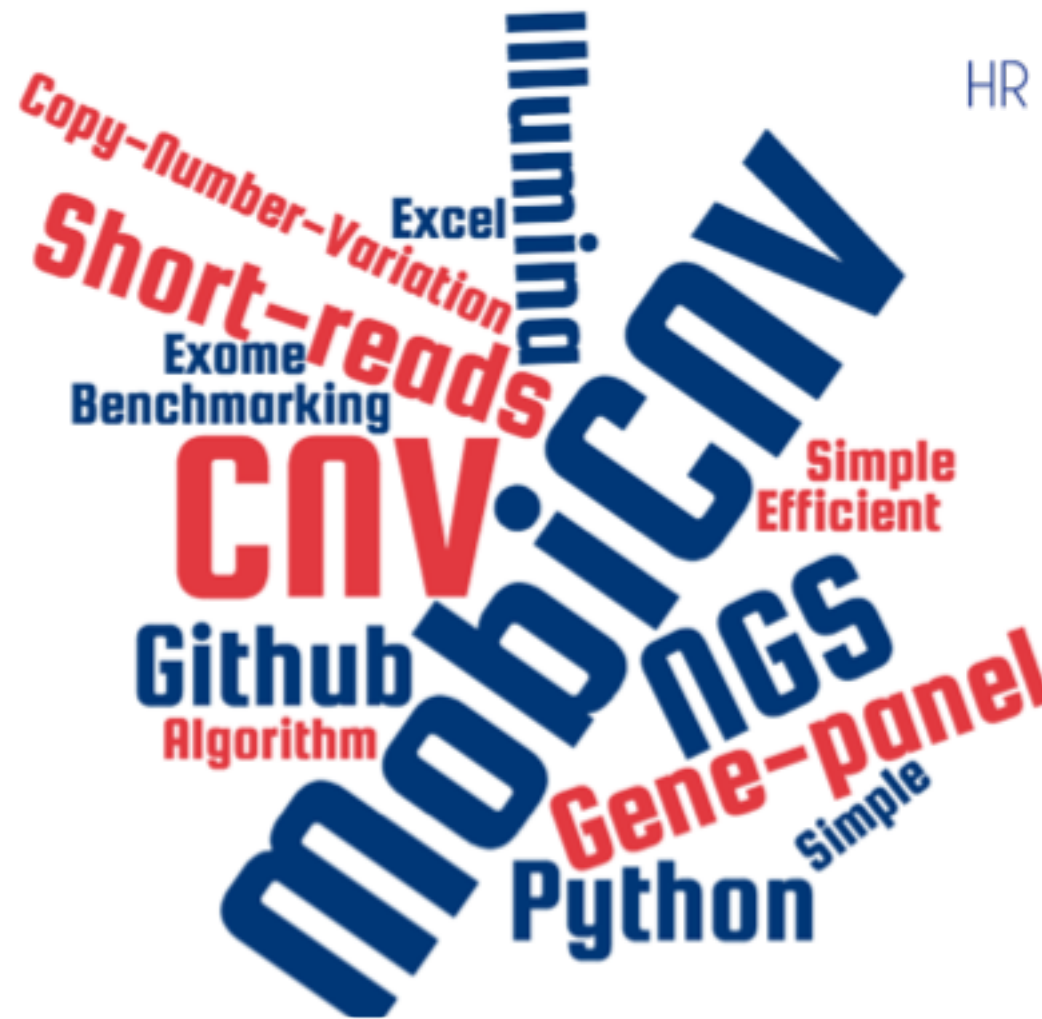
¹ Laboratoire de génétique des maladies rares, EA7402, Université de Montpellier, Laboratoire de génétique moléculaire, CHU de Montpellier, Montpellier France
² Laboratoire de biopathologie cellulaire et tissulaire des tumeurs, CHU de Montpellier, Université de Montpellier, Montpellier France



HR EXCELLENCE IN RESEARCH

What is MobiCNV?

- A read depth based Copy Number Variants detection algorithm for NGS short reads
- Extremely simple to operate, optimised for gene panels
- Include a gender prediction module
- Can operate on a desired subset of genes
- UNIX or Microsoft Windows OS



MobiCNV raw computations

X regions, n samples
Example for sample 1



intra-sample normalisation:

$$A = \frac{\text{region DoC sample 1}}{\text{global mean DoC sample 1}}$$



inter-sample normalisation:

$$B = \frac{\text{region DoC all samples except 1}}{\text{global mean DoC all samples except 1}}$$



region ratio:

$$r = \frac{A}{B}$$



region standard deviation:

$$\sigma(x) = \sqrt{\frac{\sum_{r=1}^n (r_x - m)^2}{n}}$$

x: region considered
n: number of samples
r: ratio for region x for each sample, e.g. $r_{x,1}$ = ratio region x for sample 1
m: mean r_x for all samples
DoC: Depth of Coverage, number of reads for a particular position in the genome

Sensitivity and specificity

MobiCNV vs. MUSKATTM (Sophia genetics CIVD CNV calling algorithm), manually curated

- 3 NGS runs (n=36), 3 000+ regions per sample

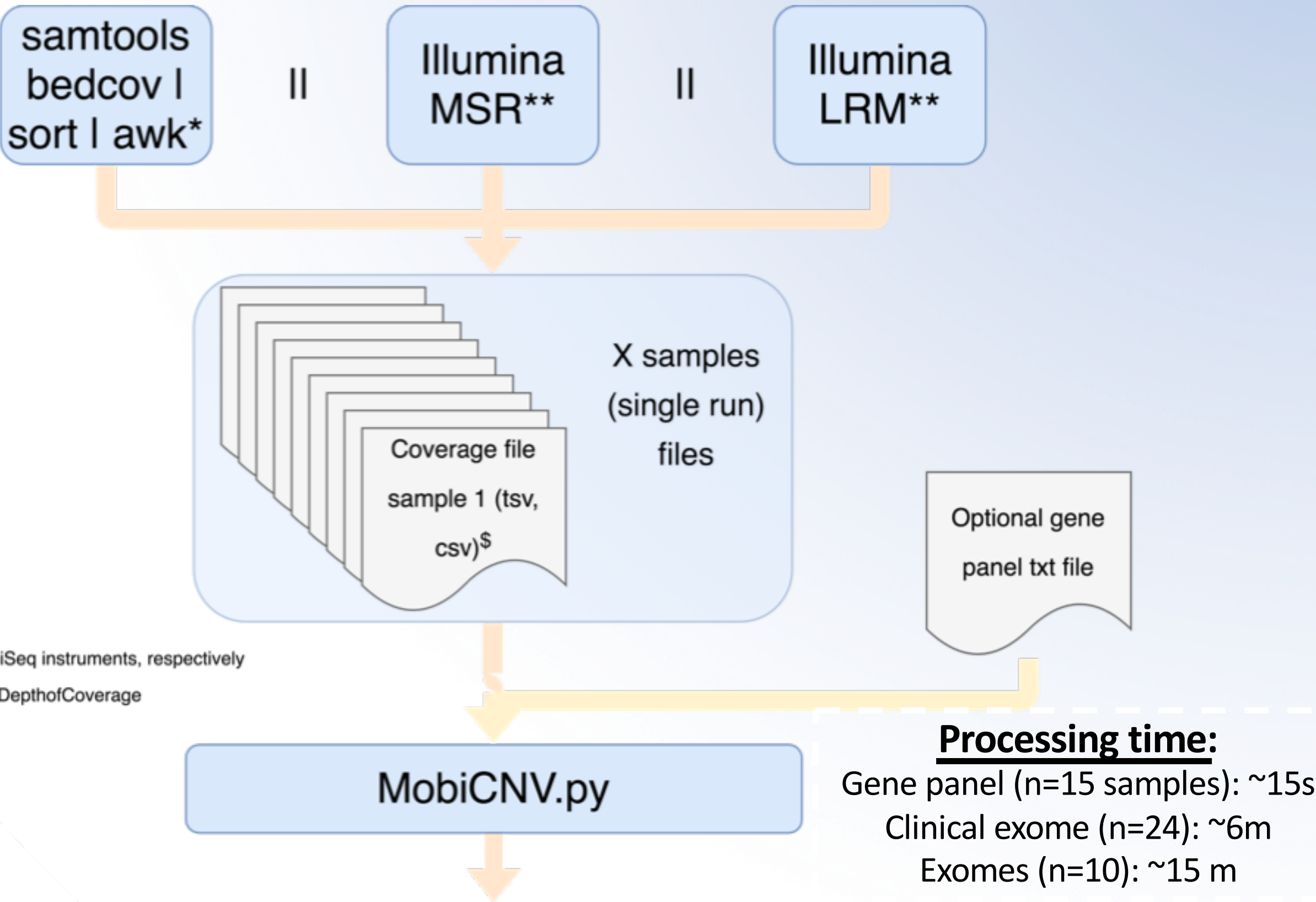
Defining heterozygosity thresholds:

	Threshold	TPR* (%)	TNR* (%)	PPV* (%)	# detected event
2 σ	0.8 and 1.2	87.09	98.53	1.64	1638
	0.75 and 1.25	83.87	99.57	5.45	477
	0.7 and 1.3	80.64	99.71	8.09	309
2.5 σ	0.8 and 1.2	77.41	99.09	2.37	1009
	0.75 and 1.25	77.41	99.58	5.22	459
	0.7 and 1.3	77.41	99.80	11.11	216
3 σ	0.8 and 1.2	35.48	99.48	1.95	562
	0.75 and 1.25	35.48	99.74	4.12	267
	0.7 and 1.3	35.48	99.87	8.87	124

*TPR: True Positive Rate; TNR: True Negative Rate; PPV: Positive Predictive Value

Best compromise between sensitivity and specificity

Computing data with MobiCNV



*from any custom pipeline
**Illumina analysis pipelines for Miseq and MiniSeq instruments, respectively
*format: chr,start,end,region_annotation,meanDepthofCoverage

Genomic coordinates | Custom annotation | Mean from tsv | depth of coverage | Samples and events: summary sheet showing 39 regions with events (15 samples, 2500+ regions)

Chromosome	Start	End	Annotation	Mean DoC	CD	CE	CF	CG	CH	CI	CJ	CK	CL	CM	CN	CO	CP	CQ	CR	CS	CT	CU	CV
chr1	6484773	6485396	ESPN.NM_031475-1	296	1	1.009	0.832	0.83	1.31	1.021	1.054	0.938	1.048	0.977	1.1	1.075	0.88	0.903	0.844				
chr1	4126434	4126521	KCNMA1.NM_004700-5	462	1.088	1.039	0.983	0.968	1.312	0.977	0.854	0.856	1.08	0.902	1.039	1.09	0.934	0.996	0.854				
chr1	100817942	100818007	CCDC18A.NM_03139210-1.1	63	1.049	0.922	0.874	0.913	1.374	0.798	0.865	1.059	0.816	0.932	1.218	1.267	0.971	0.779	1.01				
chr1	213070976	213071058	FLVCR1.NM_014053-10.4	81	0.917	1.185	1.008	0.985	1.076	1.045	0.94	0.857	1.431	0.754	1.053	1.099	0.731	0.754	1.292				
chr2	26781544	26781556	OTOF.NM_124248-1.2	156	0.949	1.159	0.91	0.98	0.976	1.252	0.887	0.886	0.957	1.083	0.988	1.252	1.083	1	0.902				
chr2	27603578	27603681	ZNF513.NM_144631-1	252	1.13	1.323	0.834	0.918	1.28	1.078	0.937	0.953	0.985	0.76	1.185	1.005	0.758	0.906	0.865				
chr4	6271496	6271822	WFS1.NM_006005-1	290	1.135	1.163	0.932	0.822	1.318	0.949	0.858	1.008	1.062	0.949	0.994	1.09	0.939	0.909	0.822				
chr4	48012815	48012897	CACNA1C.NM_000907-8	40	1	0.881	1.246	1.33	0.995	0.636	1.123	0.988	1.085	0.627	0.909	0.934	1.121	1.244	1.277				
chr4	8853725	88537455	OSPP.NM_014208-5.7	187	0.799	1.163	1.007	0.613	1.146	1.086	1.099	0.637	1.016	0.805	1.281	1.099	0.997	1.066	1.049				
chr9	140094688	140095244	TPRN.NM_01128228-1.2	50	0.964	1.21	0.904	0.714	1.548	0.94	1.037	0.94	0.759	1.061	1.134	0.934	0.855	0.798	0.936				
chr10	73555816	73555923	COH2.NM_00171933-1.2	185	1.288	1.003	1.093	0.849	1.141	0.972	0.9	0.885	1.007	0.901	1.103	1.08	0.99	0.934	0.775				
chr11	61722497	61722571	BEST1.NM_004183-3.1	326	0.985	0.905	1.017	1.021	1.308	0.919	1.004	0.812	0.816	1.04	1.135	1.04	0.853	0.953	1.013				
chr11	67289348	67289517	CARPY.NM_018366-3	261	1.095	1.358	0.958	1.04	1.002	0.979	0.872	0.796	1.132	0.86	1.131	0.956	0.865	0.984	0.988				
chr11	76858742	76859070	MYOT.NM_000260-4	356	1.009	1.095	0.902	1.012	1.301	0.966	0.93	0.89	0.976	0.947	1.024	1.048	0.95	0.93	0.904				
chr11	76870417	76870534	MYOT.NM_000260-10	453	0.993	0.953	0.989	0.927	1.321	0.868	0.937	0.852	0.862	0.907	1.127	1.142	1	0.902	0.999				
chr11	110167111	110167499	ROCK1.NM_00120492-1	41	1.088	1.164	1.029	0.913	1.059	0.913	0.771	1.059	0.609	1.147	0.985	1.059	0.971	0.928	1.179				
chr12	15794537	15794537	EPB1.NM_004471-1.1	527	1.021	0.773	0.887	0.898	0.878	0.885	1.021	1.302	0.87	1.097	0.951	1	0.981	1.043	1.203				
chr12	100815331	100815429	SLC17A8.NM_139319-12.4	585	0.986	0.822	0.984	1.193	0.835	1.184	0.954	1.388	0.983	0.95	0.837	0.923	1.048	1.052	1.187				
chr15	43903038	43903181	STREC.NM_153700-14	511	0.946	1.194	0.807	0.752	1.137	1.142	0.787	1.052	1.762	0.959	1.397	0.422	0.77	0.726	1.499				
chr15	43903033	43903061	STREC.NM_153700-12	448	0.847	1.081	1.218	1.158	0.955	0.94	0.95	0.973	0.844	0.676	0.87	1.091	1.155	1.26	0.856				
chr15	83621210	83621556	HOMER2.NM_199330-1	17	1	1.214	0.828	0.966	1.071	1.321	0.793	1.5	0.793	1.143	0.897	0.69	1.143	1	0.621				
chr16	2525076	2525244	TRC1D2.NM_00199107-1	17	1.276	1.103	1.393	0.897	1.429	1.034	0.767	0.767	1	0.65	0.897	1.138	1.034	0.862	0.862				
chr16	21742073	21742138	OTOA.NM_144672-18	622	1.046	1.201	0.82	0.907	1.122	0.837	1.275	0.607	1.116	0.835	1.066	0.839	0.815	1.137	1.198				
chr16	2175968	2175974	OTOA.NM_144672-22	905	0.87	1.131	0.924	1.04	1.08	0.919	1.077	1.024	1.065	1.065	0.881	1.059	1.01	1.035	0.984				
chr16	21756128	21756436	OTOA.NM_144672-23	899	0.884	1.029	0.814	1.049	1.013	0.912	1.075	1.082	1.03	1.082	0.927	1.08	1.115	0.969	1.126				
chr17	21771718	21772058	OTOA.NM_144672-18	520	1.017	1.043	0.867	0.938	0.845	0.883	1.043	0.883	1.04	0.942	0.887	0.978	1.008						
chr17	79479098	79479464	ACTG1.NM_00199954-2:ACTG	472	1.168	1.097	1.004	0.893	1.305	0.916	0.912	0.948	0.882	0.829	1.129	1.039	0.943	0.913	0.854				
chr17	79479098	79479464	ACTG1.NM_00199954-1	222	1.12	1.072	0.918	0.829	1.418	1.01	0.78	0.926	0.849	0.889	1.188	0.956	0.934	0.839	0.907				
chr20	60813467	60813736	ORP2L.NM_014895-1	79	1.213	1	1.033	0.877	1.631	1.117	0.794	0.931	0.907	0.823	0.848	1.131	0.786	0.823	0.931				
chr22	36717728	36717933	MYH9.NM_002473-7	715	1.078	1.034	0.974	0.99	1.129	0.985	0.91	1.335	0.967	1.046	0.904	0.945	0.93	0.916					
chr22	36783760	36784196	MYH9.NM_002473-1	297	1.056	1.083	0.91	0.843	1.235	0.883	0.824	0.992	0.938	1.008	0.996	1.004	0.973	1.019	0.91				
chr22	38142157	38142225	TRIOBP.NM_138632-1.1	148	0.9	1.307	0.951	0.728	1.350	0.983	0.809	0.909	0.713	1.021	1.058	1.129	0.902	1.033	0.996				
chr22	38142232	38142238	TRIOBP.NM_138632-1.2	109	0.905	1.034	1.006	0.676	1.282	0.989	0.845	1.079	0.672	1.205	0.989	1.124	0.933	1.056	1.039				
chr22	38142332	38142501	TRIOBP.NM_138632-1.3	112	0.891	0.908	1.055	0.875	1.428	0.975	0.902	1.318	1.033	0.967	1.071	1.066	0.737	0.87	0.943				
chrX	38150136	38150283	RYR2.NM_000328-13	545	0.888	0.892	1.041	1.016	0.899	1.109	1.054	1.05	0.837	0.849	0.91	0.959	1.041	1.124	1.381				
chrX	38186511	38186858	RYR2.NM_000328-1	330	1.142	0.959	1.008	1.354	0.898	0.946	0.829	1.108	0.98	0.985	1.043	1.046	0.905	0.868					
chrX	82764719	82764793	POU3F4.NM_000307-1.2	239	0.849	0.899	1.015	1.088	0.969	1.008	1.424	0.687	1.273	0.906	1.106	1.017	0.936	1.092	0.981				

Legend

Gender prediction

Perspectives & Developments:

Increase statistical power:
Resampling with Monte Carlo Markov Chain

Recode MobiCNV in parallelisable friendly language (D, Scala...)

Take it home!

Poster

Website

Github

